

Teacher Guide — middle / module-11-multiplication-near-base

IKS Curriculum

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Overview

Module 11 — Multiplication of Numbers Close to a Power of 10 (Nikhilam Method)

Age band: Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min

Learning objectives

By the end of this module, students will be able to:

1. **Recall** the Nikhilam procedure for multiplying two numbers close to the same base (10, 100, 1000): write each as deficit/excess from the base, cross-subtract or cross-add for the “left part,” multiply the deficits for the “right part,” and read off the answer.
2. **Execute** the method on 2-digit × 2-digit problems near 100 (both below, both above, and mixed) in under 10 seconds mentally.
3. **Extend** the same method to 3-digit × 3-digit problems near 1000, padding the right part to the correct number of digits.
4. **Explain** why the method works using the algebraic identity $(B - a)(B - b) = B(B - a - b) + ab$.
5. **Decide** when the Nikhilam method is faster than long multiplication, and when long multiplication is the better choice.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems (Tirthaji’s sūtras) with mainstream arithmetic.
- **§4.23** — Experiential learning via paired drills and timed mental-math sprints.
- **§5.1** — Foundational numeracy and computational fluency at upper-primary / middle-school level.
- **§11.8** — Holistic, multidisciplinary linkage: algebra (proof) meets arithmetic (technique).

Prerequisites

- Multiplication tables fluent up to 10×10 .
- Standard long-multiplication algorithm at 2-digit $\times$ 2-digit.
- Comfort with negative numbers at the level of $7 - 9 = -2$.
- Familiarity with place value at hundreds and thousands.
- Module 9 (Subtraction by Complements) recommended but not required — the same Nikhilam sūtra is used.

Key vocabulary

See `_shared/glossary.md`.

- *Nikhilam Navataścaramaṁ Daśataḥ* (निखिलं नवतश्चरमं दशतः) — “all from nine and the last from ten” — Tirthaji’s 2nd sūtra; the rule for computing complements from a power of 10.
- *Ūrdhva-tiryagbhyām* (ऊर्ध्व-तिर्यग्भ्याम्) — “vertically and crosswise” — Tirthaji’s 3rd sūtra; the general multiplication sūtra. Compared with Nikhilam on Day 7.
- *deficit / excess* — how much a number falls short of (or exceeds) the base.
- *base* — a power of 10 (10, 100, 1000) the two numbers are close to.
- *sūtra* — short aphorism / formula.

Materials checklist

- Whiteboard with a vertical line drawn down the middle (the “Nikhilam split”).
- Pre-printed problem sheets, graded from “both below base” to “mixed sign.”
- Stopwatch / phone timer per pair.
- A class wall-poster: the 4-line Nikhilam template (see Day 1).
- Notebook and pen per student.

Day-by-day overview

Day	Title	Focus	Assessment
1	The Hook — 97×96 in three seconds	Baseline timed test; teacher demo	Self-timed baseline
2	The Setup — deficits from 10	Single-digit $\times$ single-digit (6–9)	Pattern quiz
3	Both Below 100	2-digit $\times$ 2-digit, both close to 100	Formative quiz Part A
4	Both Above 100	Excess instead of deficit	Formative quiz Part B
5	Mixed Cases	One above, one below — sign care	<code>activities/activity-01-mixed-signs.md</code>
6	Near 1000 —	3-digit $\times$ 3-digit	Formative quiz Part C

Day	Title	Focus	Assessment
	padding the right part		
7	Why It Works — the algebra	$(B-a)(B-b) = B(B-a-b) + ab$	Reflection paragraph
8	When To Use It — vs. long multiplication	Decide-or-don't game	Decision quiz
9	Speed Sprints + Project	Mental-math sprint; project starts	Project draft
10	Final Assessment + Showcase	Summative quiz + project share	Summative + project

Citation sources used

- **SutraGanita corpus** (/Users/sree/Projects/SutraGanita/content/) — primary references in `sources.md`:
 - *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — sections on Nikhilam-based multiplication near a base.
 - *Microsoft Word - sutras1.pdf* — list of 16 Tirthaji sūtras (Nikhilam = sūtra #2).
 - *Vedic_Maths.pdf* (ISKCON Desire Tree) — practice problem sets.

Project brief (one-line teaser)

Each pair makes a 60-second teaching video, a printed 20-problem workbook, OR a comparative speed-test data project demonstrating Nikhilam multiplication near a base. Full brief in `assessments/project-brief.md`.

Honest framing in one paragraph

The Nikhilam-based “multiplication near a base” technique appears in Bharati Krishna Tirthaji’s *Vedic Mathematics* (1965, posthumous). The mathematics is genuine, fast, and useful when both numbers are within ~5–10 of a power of 10. The *attribution* — that this came from an *Atharva-veda Pariśiṣṭa* — is contested by mainstream historians of Indian mathematics (Dani 1993, Plofker 2009); see `teacher-notes.md` for how to frame this for Middle-band students.

10-Day Lesson Plan

Module 11 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Hook — 97×96	days/day-01-	Baseline timed test

Day	Session title	Lesson file	Activity / assessment
	in three seconds	hook.md	
2	The Setup — deficits from 10	days/day-02-deficits-from-ten.md	Pattern recall + 2-min sprint
3	Both Below 100	days/day-03-both-below-100.md	quizzes/formative.md Part A
4	Both Above 100	days/day-04-both-above-100.md	quizzes/formative.md Part B
5	Mixed Cases (one above, one below)	days/day-05-mixed-cases.md	activities/activity-01-mixed-signs.md
6	Near 1000 — padding the right part	days/day-06-near-1000.md	quizzes/formative.md Part C
7	Why It Works — the algebra	days/day-07-why-it-works.md	Reflection paragraph
8	When To Use It — vs. long multiplication	days/day-08-when-to-use.md	Decision quiz
9	Speed Sprints + Project Start	days/day-09-speed-and-project.md	activities/activity-02-speed-sprint.md
10	Final Assessment + Showcase	days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- Day 1’s baseline (timing each student on 97×96 solved by long multiplication) is the motivational anchor. Day 9 retests the same problem set under the Nikhilam method; the visible speedup is the module’s payoff.
- Days 2–6 are the core. If the class catches on fast, you can combine Days 3+4 (both-below + both-above) into one period and gain a day for Day 9’s sprint.
- Day 7 (the algebra) is non-negotiable for Middle band — the *why* prevents students from treating the method as a magic trick.
- Day 8 (“when to use it”) balances enthusiasm with honesty: this method is only fast in a narrow regime. Students should walk away knowing when to switch back to long multiplication.

Daily warm-up (recurring)

Every class starts with **1 minute of “complement to 10” fluency**: teacher calls out a single digit; students call back the deficit (e.g. teacher says “7,” class chants “3”). This is the same warm-up as Module 5 / Module 9, reused — it builds the foundational reflex.

Daily warm-up (Day 4 onward)

Add a **1-minute “complement to 100”** round: teacher says “97,” class chants “3.” Teacher says “88,” class chants “12.” This builds the deficit-from-100 reflex that drives 2-digit Nikhilam.

What students leave the module with

- Mental Nikhilam multiplication of 2-digit numbers near 100 in under 5 seconds.
- A clear sense of when this method is faster than long multiplication, and when it is not.
- Algebraic understanding: they can prove the method with one line of algebra.
- Awareness that “Vedic mathematics” is a 20th-century pedagogical project, not strictly ancient — see `teacher-notes.md`.
- One artefact (their project) they can share with younger students or family.

Teacher Notes

Teacher Notes — Module 11 Multiplication Near a Base (Middle)

Posture

This module trades on a specific, narrow win: when both factors are within ~5–10 of a power of 10, the Nikhilam method beats long multiplication by a wide margin. *Outside that regime, it loses.* Frame this honestly from Day 1. The point is not “Vedic math beats Western math” — both are useful in different regimes.

The Sanskrit sūtra name (*Nikhilam Navataścaramaṁ Daśataḥ*) is the cultural anchor. The *technique* is what students take home. Repeat the name often enough that students know what it sounds like by Day 6; don’t quiz them on the Sanskrit until Day 10 (and even then it’s a 1-mark question).

Honest framing in three sentences

1. The *Nikhilam* sūtra name is genuine — it appears in Bharati Krishna Tirthaji’s *Vedic Mathematics* (1965, posthumous; manuscript dated 1911–1918).
2. Tirthaji claimed the sūtra came from an *Atharva-veda Pariśiṣṭa* (Vedic appendix-text); **no other scholar has produced this manuscript**, and S. G. Dani (1993) argues the sūtras are 20th-century synthesis rather than ancient.
3. The mathematics is genuine and pedagogically valuable. The provenance is contested. At Middle band, just be honest about both; at Senior band, students read the debate.

Avoid claims like “ancient Indians knew this 3000 years ago.” There is no scholarly evidence for that specific claim about this specific technique.

Common misconceptions

1. **“This works for any multiplication.”** No. It works *only* when both numbers are close to the same power of 10. 47×53 (not near 50, not near 100) loses every advantage. Day 8 handles this honestly.
2. **“The right part is just deficit $\times$ deficit, no rules.”** Wrong — the right part must be padded to match the number of zeros in the base. For base 100, the right part takes *two* digits (e.g. $97 \times 96 \rightarrow 3 \times 4 = 12 \rightarrow '12'$); for $99 \times 98 \rightarrow 1 \times 2 = 02$, not 2. For base 1000, three digits.
3. **“If the right part overflows ($\geq$ base), I’m stuck.”** Handle by carrying. For 88×89 : deficits 12 and 11; right part = 132 (3 digits, but base is 100 $\rightarrow$ 2-digit pad). Carry the 1 to the left part: left = $(88 - 11) = 77 \rightarrow 77 + 1 = 78$; right = 32. Answer: 7832. Day 4 / Day 5 covers this gotcha.
4. **“In the mixed case (one above, one below) the right part is just deficit $\times$ excess.”** Sign matters. For 102×98 : excess +2, deficit -2. Right part = $(+2)(-2) = -4$. The “left” computed by either cross gives 100. So: $100 \times 100 + (-4) = 10000 - 4 = 9996$. Day 5 handles the sign care.
5. **“The dot/Nikhilam is the same as the long-multiplication carry.”** Conceptually related but procedurally different. The deficit-and-cross is what makes it fast. Don’t blur the methods on Day 1; introduce Nikhilam cleanly first.

Differentiation hints

- **Tier up:** Move to base 50 (a “working base”). For 48×47 , deficits from 50 are 2 and 3. Cross-subtract: $48 - 3 = 45$. Multiply: $45 \times 50 = 2250$. Add the deficit product: 6. Answer 2256. Tirthaji calls this *sub-base* multiplication; sūtra *Ānurūpyeṇa*. Optional enrichment for fast finishers.
- **Tier down:** Stay with single-digit near-10 (7×8 , 6×9) for the full first week. Don’t push 2-digit until the student is fluent at the 1-digit case.

Differential pacing

- A strong class can compress to 7 days by combining Days 3+4 and Days 5+6.
- A class shaky on multiplication tables should NOT skip Day 2. The “complement from 10” reflex is the prerequisite. Pull strugglers into extra drill at break.

Sensitivity notes

- Timed mental-math activities can spike anxiety. Frame them as **personal baselines** — students race themselves over the module, not each other. No public leaderboards.
- Some families teach “Vedic mathematics” with religious / patriotic pride. The honest historical framing may feel diminishing. The teacher line: “*The method is a real Indian achievement. We just don’t oversell its age.*” If a parent objects, point them to the parent guide and offer to talk after class.
- Avoid framing this as “Indian math vs Western math.” Both are useful. The standard long-multiplication algorithm is itself of mixed origin (al-Khwārizmī built on Brahmagupta).

Common arithmetic errors to watch for

- **Forgetting to pad the right part.** $99 \times 98 \rightarrow$ right part = 2, but it should be 02. Answer should be 9702, not 972.
- **Forgetting to carry when the right part overflows.** $88 \times 89 \rightarrow$ right part 132; needs to carry 1 to the left.
- **Sign error in mixed case.** Most students will try to write a negative right part directly. Show them the “borrow from the left” maneuver: $102 \times 98 \rightarrow$ left = 100, right = -4 $\rightarrow$ borrow 1 from left $\times 100 = 100$; $(99 \times 100) + (100 - 4) = 9900 + 96 = 9996$. Or compute directly: $(100)(100) - 4 = 9996$.
- **Confusion between base 10 and base 100.** For 9×8 (base 10), pad to 1 digit. For 99×98 (base 100), pad to 2 digits. Reinforce: number of zeros in base = number of digits in the padded right part.

Project oversight

- Three project formats (60-sec teaching video, 20-problem workbook, comparative speed-test data project) each have different production challenges. See Module 5’s notes for the cross-cutting issues — they apply here too.
- For the data project, students need a partner / family member who’ll let themselves be timed on long multiplication. Confirm by Day 9 that this is actually arranged.

Citation discipline

Citations for this module name **both** the Sanskrit sūtra (*Nikhilam Navataścaramaṁ Daśataḥ*) AND the modern source (e.g. *Vedic Mathematics Batch1.pdf*). Student projects should cite at minimum the modern PDF; mentioning the sūtra by name is encouraged but optional.

If you only have 6 days

- Day 1: hook + baseline + Day 2 (deficit setup) combined.
- Day 2: Day 3 (both below 100) — the core case.
- Day 3: Day 5 (mixed cases) — covers Day 4 incidentally.
- Day 4: Day 7 (algebra) + Day 8 (when to use it) combined.
- Day 5: project work.
- Day 6: assessment + showcase.

Drop Day 6 (near 1000) and Day 9 (sprint) if pressed; preserve the algebra (Day 7) and the “when to use” discussion (Day 8).

Sources

Sources — Module 11 Multiplication Near a Base (Middle)

The **vidya-karana RAG** does not cover Vedic-math sources, so this module draws directly from SutraGanita’s source corpus (/Users/sree/Projects/SutraGanita/content/). All PDFs cited below are accessible in that directory.

Primary source (used most heavily)

Vedic Mathematics Batch1.pdf — *Vedic Cultural Center (ISKCON), 1420 228th Ave SE, Sammamish WA 98075*

- Sections on Nikhilam-based multiplication near a base — both numbers below (the 7×8 , 97×96 family), both numbers above (103×104), and the mixed case (102×98).
- The cross-subtraction / cross-addition for the left part.
- The deficit $\times$ deficit product for the right part and the “two-digit padding when base = 100” rule.

Cite as: *Vedic Mathematics Batch1.pdf* (Nikhilam multiplication section).

Supporting sources

- **Microsoft Word - sutras1.pdf** — single page listing all 16 Tirthaji sūtras and their 13 sub-sūtras. Used to name *Nikhilam Navataścaramaṁ Daśataḥ* (sūtra #2) and *Ūrdhva-tiryagbhyām* (sūtra #3, mentioned on Day 8 for contrast).
- **Vedic_Maths.pdf** (ISKCON Desire Tree) — workbook style; contains practice problem sets for multiplication near a base. Used for drills in Days 3–6.
- **FUNDAMENTAL AND VEDIC MATHEMATICS .pdf** — comprehensive 65-page treatment; useful as a teacher reference for the algebraic proof on Day 7.

Historical context

- **Tirthaji, Bharati Krishna.** *Vedic Mathematics: Sixteen Mathematical Formulae from the Vedas* (1965), Motilal Banarsidass. — Tirthaji’s foundational text. The Nikhilam multiplication scheme is one of his showcase examples.
- **Plofker, Kim.** *Mathematics in India* (2009), Princeton University Press. — Historical context of Indian mathematics; useful for situating Tirthaji’s work in the broader history.
- **Dani, S. G.** “Myths and Reality: On Vedic Mathematics” (1993, *Frontline*; reprinted in *Resonance* 2001). — Standard scholarly critique. At Middle band this is a teacher reference, not student reading; at Senior band it is required reading.

Cross-references within the curriculum

- **Module 5 — Bindu Paddhati (Dot Addition).** Uses the *Sankalana Vyavakalanābhyām* “completion to a base” idea. Same family of techniques.

- **Module 7 — Multiplication by 11.** A special-case shortcut at a *different* multiplier. The contrast ($\times 11$ always works; Nikhilam only works near a base) is worth a 30-sec mention on Day 8.
- **Module 9 — Subtraction by Complements.** Uses *the same* sūtra (*Nikhilam Navataścaramaṁ Daśataḥ*) for finding complements. If students have done Module 9, the “deficit from base” idea is already familiar — skip the long explanation on Day 2 and double drills.

What to tell students about the citation chain

At Middle band, the citation pattern is: “the *Nikhilam* method appears in Tirthaji’s *Vedic Mathematics* (1965). The procedural details we use are from *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish).” Both the sūtra name (Sanskrit) AND the modern source (PDF) get cited.

A one-line honest framing for student-facing material: “*This method is published in Tirthaji’s 1965 book. The Sanskrit name of the sūtra is real; the claim that the sūtra itself comes from an ancient Vedic text is debated by historians.*” Senior-band students will engage with this debate in detail.

Day 1 — The Hook: 97×96 in Three Seconds

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will record a personal baseline time for 97×96 solved by long multiplication, and will have seen the *Nikhilam* method produce the same answer in under 5 seconds.

Materials

- Printed baseline sheet: a single problem $97 \times 96 = ?$ with workspace and “your time” line. One per student.
- Stopwatch or wall clock with a second hand. One per pair is enough.
- Whiteboard with a single vertical dividing line drawn down the middle (the “Nikhilam split”).

Vocabulary introduced today

- *Nikhilam* (IAST: nikhilam; lit. “all” — short for *Nikhilam Navataścaramaṁ Daśataḥ*, “all from 9 and the last from 10”) — Tirthaji’s 2nd sūtra; here, the rule for multiplication near a base.
- *deficit* — how much a number falls short of the base (97 is 3 short of 100; deficit = 3).
- *base* — a power of 10 the two numbers are close to (here, 100).

Lesson flow

Warm-up (5 min)

- Teacher: “How long does it take you to multiply 97×96 ?” Take 3–4 guesses. Write the range (e.g. “15 sec to 1 min”).
- Teacher: “I’ll show you a way to do it in under 5 seconds. Then we’ll talk about why.”

Baseline timing (10 min)

- Hand out the baseline sheet face-down.
- Students write their name and predicted time on the back.
- On “Go,” flip and solve 97×96 using long multiplication.
- Record actual time. Mark answer; honest self-correction. (Correct answer: 9312.)
- Teacher walks around. Most will take 30–60 seconds. A few faster, a few slower.

Teacher demonstration (10 min)

- On the board, write the problem:

$$\begin{array}{r} 97 \\ \times 96 \\ \hline \end{array}$$

- Step 1: Find the deficits from 100.** Write -3 next to 97 and -4 next to 96.

$$\begin{array}{r|l} 97 & -3 \\ 96 & -4 \end{array}$$

- Step 2: Cross-subtract for the LEFT part.** “Take one number minus the *other* number’s deficit. $97 - 4 = 93$. (Equivalently: $96 - 3 = 93$. Either cross gives the same answer.)”
- Step 3: Multiply the deficits for the RIGHT part.** “ $3 \times 4 = 12$.”
- Step 4: Stick them together.** “93 on the left, 12 on the right $\rightarrow 9312$.”

$$\begin{array}{r|l} 97 & -3 \\ 96 & -4 \\ \hline 93 & 12 \end{array} \rightarrow 9312$$

- Verify: long-multiplication gives 9312. ✓
- “I never multiplied any 2-digit number by another 2-digit number. I only did $97 - 4$ and 3×4 . That’s the entire trick — when both numbers are near a base.”

Activity (15 min) — First try

- Hand each student a second sheet with four problems:
 - 98×97 , 96×95 , 99×99 , 97×92 .

- Students try the Nikhilam method on each, then verify with long multiplication.
- Time themselves. Expect mistakes and slow times on the first try.

Wrap-up (5 min)

- Show of hands: who got at least one right with Nikhilam? Who was *faster* with Nikhilam already? (Most won't be — yet.)
- Preview Day 2: “Tomorrow we'll work on the foundational skill — instantly finding the deficit of any number from 10 or 100.”

Student-facing content

The Nikhilam Method — Multiplication Near a Base

When both numbers are near a power of 10 (10, 100, 1000):

1. **Find the deficit** of each from the base.
2. **Cross-subtract** for the LEFT part (one number minus the *other's* deficit).
3. **Multiply the deficits** for the RIGHT part.
4. **Concatenate**: LEFT | RIGHT = the answer.

Today's baseline: my time on 97×96 using long multiplication = ____ seconds. My time using Nikhilam (first try) = ____ seconds.

I expect this gap to FLIP by Day 9.

Homework

- Practice the “complement to 10” reflex: write the deficit of each from 10.
 - $7 \rightarrow , 8 \rightarrow , 9 \rightarrow , 6 \rightarrow , 5 \rightarrow , 4 \rightarrow , 3 \rightarrow , 2 \rightarrow , 1 \rightarrow \underline{\quad}$.
- Do all 9 in under 10 seconds.

Differentiation

- **Tier up:** Try 103×104 (both *above* 100). Note: instead of cross-subtract, you'll need to cross-add. See if you can work out the pattern before Day 4.
- **Tier down:** If 2-digit feels too fast, drop to single-digit near 10 (7×8 , 6×9). We'll formally do this on Day 2.

Teacher notes

- The baseline time is the *single most important* artefact of Day 1. Make sure every student records it honestly. Day 9's retest depends on it.
- Don't promise a specific speedup. Say “you'll likely be faster by Day 9; how much depends on you.”
- Some students will ask “why does this work?” — defer to Day 7. “Great question. We'll prove it. For now, just notice that it does work.”
- A few students will try to short-cut and use -3×-4 as the right part, getting a sign-error. Wait until Day 5 for the mixed case. For today, all deficits are positive.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication section.
- *Microsoft Word - sutras1.pdf* — sūtra #2: *Nikhilam Navataścaramaṁ Daśataḥ*.

Day 2 — The Setup: Deficits from 10 (Single-Digit × Single-Digit)

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will instantly state the deficit of any single digit from 10, AND will solve any single-digit × single-digit multiplication where both digits are 6–9 using the Nikhilam method.

Materials

- Class wall-poster: “Deficits from 10” — the table $1 \rightarrow 9, 2 \rightarrow 8, \dots, 9 \rightarrow 1$. Display before class.
- Drill sheet: 20 single-digit × single-digit problems (both factors 6–9).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-10 lightning round.** Teacher calls 10 single digits in 30 seconds; class chants the deficit. ($7 \rightarrow 3, 8 \rightarrow 2, 6 \rightarrow 4, 9 \rightarrow 1, \dots$)
- Repeat with a different sequence. By the end most students should be calling out the deficit faster than they can think it.

Core (15 min) — Nikhilam at the single-digit level

1. **Set up the template on the board** (use the vertical line):

$$\begin{array}{r|l} 7 & -3 \\ \times 8 & -2 \\ \hline ? & ? \end{array} \quad \begin{array}{l} \leftarrow 7 \text{ is } 3 \text{ below } 10 \\ \leftarrow 8 \text{ is } 2 \text{ below } 10 \end{array}$$

2. **The two operations:**

- LEFT part: cross-subtract. $7 - 2 = 5$. (Or $8 - 3 = 5$. Either cross gives the same number — that’s a built-in check.)
- RIGHT part: multiply the deficits. $3 \times 2 = 6$.
- Concatenate: **56**.

3. **Verify.** $7 \times 8 = 56$. ✓
4. **Why does either cross give the same answer?** Quick proof on board: $7 - 2 = 7 - (10 - 8) = 7 + 8 - 10$ $8 - 3 = 8 - (10 - 7) = 7 + 8 - 10$ Same thing. Don't dwell — Day 7 has the full algebra.
5. **The hand-trick (optional, fun).** Hold up both hands. For 7×8 : fold down 2 fingers on the left hand (deficit of 7 from 10 = 3, so keep 7 fingers up; or fold down $7 - 5 = 2$ on the left) and 3 on the right (deficit of 8 = 2, so fold down $2 + 3 = \dots$ use whichever finger-mnemonic the class likes). The number of fingers DOWN $\times 10$ = the tens; the number of fingers UP on each hand multiplied = the units. (This is a folk technique that maps to the Nikhilam algebra.) Show once, don't drill it.
6. **Worked examples in chorus:**
 - 9×8 : deficits 1, 2. Left: $9 - 2 = 7$. Right: $1 \times 2 = 2$. Answer: **72**. ✓
 - 6×9 : deficits 4, 1. Left: $6 - 1 = 5$. Right: $4 \times 1 = 4$. Answer: **54**. ✓
 - 6×7 : deficits 4, 3. Left: $6 - 3 = 3$. Right: $4 \times 3 = 12$. Hmm — right part is two digits! Carry the 1 to the left: left = $3 + 1 = 4$; right = 2. Answer: **42**. ✓
7. **Note the carry rule.** When the right part has more digits than the base has zeros (here: base 10, so right part should be 1 digit; if it's 2 digits, carry the extra). Write on board:

“If the right part overflows the base, carry the overflow into the left part.”

Activity (20 min) — Drill

Pair drills.

- Hand out the drill sheet — 20 problems, all single-digit $\times$ single-digit, both factors in 6–9.
- Working in pairs:
 - Partner A reads problem 1 aloud.
 - Partner B answers in under 3 seconds using Nikhilam.
 - Swap.
- 10 min of paired drill. Then 5 min of individual silent practice.

After 5 minutes of paired drills, do a class-level **timing test**: project 10 problems on the board; students answer all 10 individually in 60 seconds.

Wrap-up (5 min)

- Quick check: chorus answers — 7×7 ? (49.) 6×8 ? (48.) 9×9 ? (81.) 8×6 ? (48.)
- Preview Day 3: “Tomorrow — both numbers near 100. Same trick, slightly bigger.”

Student-facing content

Single-digit Nikhilam (base = 10)

1. Find each digit's **deficit from 10**.
2. Cross-subtract for the LEFT part.

3. Multiply the deficits for the RIGHT part.
4. Stick them together. If the RIGHT part is 2 digits, **carry** the overflow.

Examples: -7×8 : $7-2=5$; $3 \times 2=6 \rightarrow 56$. -9×8 : $9-2=7$; $1 \times 2=2 \rightarrow 72$. -6×7 : $6-3=3$; $4 \times 3=12 \rightarrow$ carry 1 $\rightarrow 3+1=4$; right=2 $\rightarrow 42$.

The Sanskrit sūtra: **Nikhilam Navataścaramaṁ Daśataḥ** — “all from 9 and the last from 10.”

Homework

- Practice deficit-from-10 recall. Have a parent/sibling call out single digits 1–9; you answer the deficit. Beat 20 problems in 30 seconds.
- Solve 10 problems on the back of today’s worksheet (all single-digit $\times$ single-digit, both 6–9) using Nikhilam. Self-check against your times table.

Differentiation

- **Tier up:** Try $9 \times 9 \times 9$ (chained) or apply the method to base 20 — see if 19×18 works the same way with base 20. (Yes — deficits 1, 2 $\rightarrow$ left 17, right 2 $\rightarrow 17 \times 20 + 2 = 342$. ✓)
- **Tier down:** Stick to 8×9 , 9×9 , 7×8 for today. Memorise these specific three; we’ll widen the range tomorrow.

Teacher notes

- Some students will protest: “*But I already know $7 \times 8 = 56$ from my times tables!*” Honest answer: “*Yes — for single digits the times-table is faster. The Nikhilam method is here so we can use it on 2-digit numbers tomorrow, where the times-table doesn’t help.*”
- Watch for the carry rule on 6×7 and 6×8 . This is the most common error point.
- The “either cross gives the same” property is a built-in check students can use to catch errors. Reinforce it.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, single-digit examples.
- *Microsoft Word - sutras1.pdf* — sūtra #2: **Nikhilam Navataścaramaṁ Daśataḥ**.

Day 3 — Both Numbers Below 100

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit $\times$ 2-digit multiplication where both factors are between 90 and 99 using the Nikhilam method, in under 10 seconds.

Materials

- Whiteboard with the vertical-line template.
- Class wall-poster: “Deficits from 100” — small reference card (e.g. $97 \rightarrow 3$, $88 \rightarrow 12$).
- Pre-printed drill sheet (10 problems).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-100 lightning round.** Teacher calls 2-digit numbers in the 80s and 90s; class chants the deficit. ($97 \rightarrow 3$, $88 \rightarrow 12$, $96 \rightarrow 4$, $92 \rightarrow 8$, $99 \rightarrow 1$, $85 \rightarrow 15$, ...)
- Repeat. By the end most students should be doing this in under a second.

Core (15 min) — The 2-digit $\times$ 2-digit case, both below 100

1. **The template** (vertical line, base = 100):

$$\begin{array}{r|l} 97 & -3 \\ \times 96 & -4 \\ \hline ? & ? \end{array} \quad \leftarrow \text{deficit from 100}$$

2. **Walk through 97×96** (from Day 1, now formalised):
 - LEFT: $97 - 4 = 93$.
 - RIGHT: $3 \times 4 = 12$.
 - **Pad to 2 digits** (base 100 has 2 zeros): RIGHT = 12 (already 2 digits, no pad needed).
 - Answer: **9312**.
3. **Walk through 98×99 :**
 - Deficits: 2, 1.
 - LEFT: $98 - 1 = 97$.
 - RIGHT: $2 \times 1 = 2 \rightarrow$ pad to 2 digits $\rightarrow 02$.
 - Answer: **9702**.
4. **The padding rule (write on board, boxed):**

Pad the right part to as many digits as there are zeros in the base. Base 10 $\rightarrow$ 1 digit on the right. Base 100 $\rightarrow$ 2 digits on the right. Base 1000 $\rightarrow$ 3 digits on the right.
5. **The overflow / carry case: 88×89 .**
 - Deficits: 12, 11.
 - LEFT: $88 - 11 = 77$. (Or $89 - 12 = 77$. ✓)
 - RIGHT: $12 \times 11 = 132$.

- But the right part should be only 2 digits (base 100). 132 has 3 digits — the leading 1 overflows.
- **Carry the overflow into the left:** left = $77 + 1 = 78$; right = 32.
- Answer: **7832**. ✓ (Calculator check: $88 \times 89 = 7832$.)

6. The “use either cross” check. For 97×96 :

- Either $97 - 4 = 93$ or $96 - 3 = 93$.
- If you get *different* numbers, you’ve made an error. This is your built-in error-detector.

Activity (20 min) — Drill

Pair drill, then individual silent drill.

Drill sheet (10 problems, mixed difficulty):

- | | | | |
|-------------------|--------------------|-------------------|-------------------|
| 1. 97×96 | 2. 98×99 | 3. 95×94 | 4. 93×91 |
| 5. 99×98 | 6. 92×95 | 7. 88×89 | 8. 87×92 |
| 9. 96×96 | 10. 91×87 | | |

- **5 min individual:** solve all 10 silently. No talking.
- **5 min paired check:** swap sheets. Partner reads problem 1; the other resolves it on a fresh strip; compare. Continue.
- **10 min mixed work:** problems with both-below-100, individual silent.

Answer key (teacher): | Q | Deficits | Left | Right | Answer | |—|—|—|—|—|—| | 1 | 3, 4 | 93 | 12 | 9312 | | 2 | 2, 1 | 97 | 02 | 9702 | | 3 | 5, 6 | 89 | 30 | 8930 | | 4 | 7, 9 | 84 | 63 | 8463 | | 5 | 1, 2 | 97 | 02 | 9702 | | 6 | 8, 5 | 87 | 40 | 8740 | | 7 | 12, 11 | $77+1=78$ | 32 | 7832 | | 8 | 13, 8 | $87-13=79$ or $92-8=84...$ | (see note) | 8004 | | 9 | 4, 4 | 92 | 16 | 9216 | | 10 | 9, 13 | $91-13=78$ or $87-9=78$ | 117 → carry 1 → 79, right 17 | 7917 |

(Note Q8: $87 - 8 = 79$, OR $92 - 13 = 79$. Right = $13 \times 8 = 104$. Carry 1 → 80; right = 04. Answer: 8004.)

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see quizzes/formative.md Part A.
- Collect for grading. Hand back tomorrow.

Student-facing content

Both numbers below 100

Step	What you do
1	Find each number’s deficit from 100 .
2	Cross-subtract for the LEFT part.
3	Multiply the two deficits for the RIGHT part.
4	Pad the RIGHT part to 2 digits .

Step	What you do
5	If the RIGHT part overflows (≥ 100), carry the overflow into the LEFT. Examples: $- 97 \times 96 \rightarrow 93 \mid 12 \rightarrow \mathbf{9312}$ - $98 \times 99 \rightarrow 97 \mid 02 \rightarrow \mathbf{9702}$ - $88 \times 89 \rightarrow 77+1 \mid 32 \rightarrow \mathbf{7832}$ (right part overflowed) Sanskrit name: <i>Nikhilam Navataścaramaṁ Daśataḥ</i> — “all from 9 and the last from 10.”

Homework

- 10 problems on the back of today’s worksheet, all 2-digit $\times$ 2-digit with both factors in 85–99.
- Time yourself. Beat the class average from today.

Differentiation

- **Tier up:** Try 83×89 and 78×85 (further from 100). The method still works, but the deficits are larger and the right-part overflow is more common. See how far below 100 you can go before the method stops feeling fast.
- **Tier down:** Stick to factors in 90–99 for the rest of the week. We’ll revisit 80s on Day 5.

Teacher notes

- The overflow / carry is the single most common error point. Spend time on 88×89 and 78×85 . If students don’t fluent on the carry by end of class, schedule 5 min of catch-up at the start of Day 4.
- The “use either cross” check is a real check — encourage students to do both crosses initially. By Day 6 they’ll naturally drop the second one.
- Some students will be fast on a calculator and slow on Nikhilam. Acknowledge: “*With a calculator, calculator wins. The Nikhilam method matters when you DON’T have a calculator — exams, mental estimation, daily life.*”

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, base-100 examples.

Day 4 — Both Numbers Above 100

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit $\times$ 2-digit (or 3-digit) multiplication where both factors are slightly *above* 100 (101–112) using the Nikhilam method.

Materials

- Whiteboard with the vertical-line template.
- Day 3 formative quiz returned with marks.
- Drill sheet (10 problems with both factors above 100).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-100 round** (same as Day 3).
- Add the new round: **excess from 100**. Teacher says “103,” class says “+3.” “108” → “+8.” “112” → “+12.” “101” → “+1.”
- Hand back yesterday’s formative.

Core (15 min) — Both above 100: use excess, cross-ADD

1. **The key change:** instead of *deficit* (negative), we now have *excess* (positive). Instead of cross-subtract, we **cross-add**.
2. **The template** (vertical line, base = 100):

$$\begin{array}{r|l} 103 & +3 \\ \times 104 & +4 \\ \hline ? & ? \end{array} \quad \leftarrow \text{excess from 100}$$

3. **Walk through 103×104 :**
 - Excesses: +3, +4.
 - LEFT: $103 + 4 = 107$. (Or $104 + 3 = 107$. ✓ Either cross.)
 - RIGHT: $3 \times 4 = 12$. Pad to 2 digits: 12.
 - Answer: **10712**. ✓ (Calculator check: $103 \times 104 = 10712$.)
4. **Walk through 108×107 :**
 - Excesses: +8, +7.
 - LEFT: $108 + 7 = 115$.
 - RIGHT: $8 \times 7 = 56$. Pad to 2 digits: 56.
 - Answer: **11556**. ✓
5. **The pattern:**

Both below	Both above
Cross- subtract	Cross- add
Deficit $\times$ deficit	Excess $\times$ excess
Right part is product of POSITIVE numbers (deficits)	Right part is product of POSITIVE numbers (excesses)

In both cases the right part is a positive product. The sign issue only appears in the *mixed* case (Day 5).

6. The overflow case (rare here): 112×113 .

- Excesses: +12, +13.
- LEFT: $112 + 13 = 125$.
- RIGHT: $12 \times 13 = 156$. Right part should be 2 digits; the leading 1 overflows.
- **Carry:** left = $125 + 1 = 126$; right = 56.
- Answer: **12656**. ✓

Activity (20 min) — Drill

Drill sheet (10 problems):

- | | | |
|----------------------|---------------------|---------------------|
| 1. 103×104 | 2. 102×105 | 3. 108×107 |
| 4. 109×102 | 5. 101×106 | 6. 111×104 |
| 7. 112×113 | 8. 106×109 | 9. 103×103 |
| 10. 105×105 | | |

- 5 min individual silent.
- 5 min paired check.
- 10 min mixed: alternate problems from yesterday (both below) and today (both above) on the same drill sheet — call it the “mixed-base” sheet. Students need to first **decide** whether each is “both below” or “both above” before applying the right cross-operation.

Answer key (teacher): | Q | Sign | Cross | Left | Right | Answer | |—|—|—|—|—|—|—|
 —| 1 | both above | + | 107 | 12 | 10712 | | 2 | both above | + | 107 | 10 | 10710 | | 3 | both above | + | 115 | 56 | 11556 | | 4 | both above | + | 111 | 18 | 11118 | | 5 | both above | + | 107 | 06 | 10706 | | 6 | both above | + | 115 | 44 | 11544 | | 7 | both above | + | 125+1=126 | 56 | 12656 | | 8 | both above | + | 115 | 54 | 11554 | | 9 | both above | + | 106 | 09 | 10609 | | 10 | both above | + | 110 | 25 | 11025 |

Wrap-up (5 min)

- **Formative quiz Part B (3 min)** — see quizzes/formative.md.
- Preview Day 5: “Tomorrow — what happens when ONE number is above and ONE is below? Sign care needed.”

Student-facing content

Both numbers above 100

Step	What you do
1	Find each number’s excess above 100 (positive number).
2	Cross-ADD for the LEFT part.
3	Multiply the two excesses for the RIGHT part.
4	Pad the RIGHT part to 2 digits.

Step	What you do
5	If the RIGHT part overflows, carry the overflow into the LEFT. Examples: - $103 \times 104 \rightarrow 107 \mid 12 \rightarrow \mathbf{10712}$ - $108 \times 107 \rightarrow 115 \mid 56 \rightarrow \mathbf{11556}$ - $112 \times 113 \rightarrow 126 \mid 56 \rightarrow \mathbf{12656}$ (right part overflowed) Same sūtra, mirror operation: when deficits become excesses, subtraction becomes addition.

Homework

- 10 mixed problems (both-below AND both-above mixed). Decide first, then solve. Bring tomorrow.

Differentiation

- **Tier up:** Try 1003×1004 (base 1000). The right part should be 3 digits. Confirm $1003 \times 1004 = 1007012$.
- **Tier down:** Stay with 103×104 , 102×103 , 104×105 until comfortable.

Teacher notes

- The “decide which sign first” step on the mixed drill sheet is the most important habit-builder of this day. Reinforce: *before* you start cross-adding or cross-subtracting, **glance at both numbers** and decide whether they’re both below, both above, or split.
- A few students will try to carry into the right part from the left (instead of the other way). Walk through the place-value reasoning: a “100” overflow on the right is *one more in the next column up*, which is the left part.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, “above the base” examples.
- *Vedic_Maths.pdf* (ISKCON Desire Tree) — practice problems for above-base multiplication.

Day 5 — Mixed Cases: One Above, One Below

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit $\times$ 2-digit multiplication where one factor is *above* 100 and one is *below*, handling the negative right-part correctly.

Materials

- Whiteboard.

- Drill sheet (8 mixed-sign problems).
- activities/activity-01-mixed-signs.md worksheet.

Lesson flow

Warm-up (5 min)

- Complement-to-100 round.
- Excess-from-100 round.
- Now mix: teacher says “97,” class says “-3.” Teacher says “103,” class says “+3.” This trains the sign attention.

Core (15 min) — One above, one below

1. **Set up 102×98 .** Write on board:

$$\begin{array}{r|l}
 102 & +2 \\
 \times 98 & -2 \\
 \hline
 ? & ?
 \end{array}
 \quad
 \begin{array}{l}
 \leftarrow \text{excess} \\
 \leftarrow \text{deficit}
 \end{array}$$

2. **The left part — either cross still works:**

- $102 + (-2) = 102 - 2 = 100$.
- $98 + 2 = 100$. (Same answer ✓.)

3. **The right part — sign matters:**

- $(+2) \times (-2) = -4$.
- The right part is **negative**!

4. **What does it mean to have a “negative” right part?**

- Concatenating “100” and “-4” gives... $100 \mid -4 = 100 \times 100 + (-4) = 10000 - 4 = \mathbf{9996}$.
- Check: $102 \times 98 = 9996$. ✓

5. **The “borrow” maneuver — cleaner mental version.** Same problem:

- The negative right part of -4 says: “subtract 4 from the natural concatenation $100 \mid 00 = 10000$.”
- So $102 \times 98 = 10000 - 4 = \mathbf{9996}$. Clean.

6. **Or — the “borrow from left” form:** if students prefer always-positive right parts:

- Left = 100. Right = -4. Borrow 1 from left (which is worth 100 in the right):
- New left = $100 - 1 = 99$. New right = $100 - 4 = 96$ (now positive).
- Answer: **9996**.

7. **Worked examples in chorus:**

- 104×96 : excess +4, deficit -4. Left = 100 (either cross). Right = $+4 \times -4 = -16$. Answer: $100 \times 100 - 16 = \mathbf{9984}$. ✓

- 97×103 : deficit -3 , excess $+3$. Left = 100. Right = $-3 \times +3 = -9$. Answer: **9991**. ✓
- 108×95 : excess $+8$, deficit -5 . Left = $108 - 5 = 103$ (or $95 + 8 = 103$ ✓). Right = $+8 \times -5 = -40$. Answer: $103 \times 100 - 40 = 10300 - 40 = \mathbf{10260}$. ✓

8. The pattern (write on board):

Case	Cross	Right part sign
Both below	Subtract	+ (deficit $\times$ deficit)
Both above	Add	+ (excess $\times$ excess)
Mixed	Either form — still cancels	– (excess $\times$ deficit)

Activity (20 min) — [activities/activity-01-mixed-signs.md](#)

Brief: - 8 problems on the activity sheet, all mixed-sign. - 8 min individual silent. - 7 min paired check. - 5 min: students re-do any they got wrong, OR move to challenge problems if all correct.

Wrap-up (5 min)

- Quick chorus check: $102 \times 98?$ (9996.) $104 \times 96?$ (9984.) $108 \times 95?$ (10260.)
- Preview Day 6: “*Tomorrow — base 1000. The pad becomes 3 digits.*”

Student-facing content

Mixed case: one above, one below

1. Find the **excess** of the bigger number and the **deficit** of the smaller (or just signed: + for above 100, – for below).
2. LEFT: cross-subtract using signs. $102 + (-2) = 100$. Either cross still works.
3. RIGHT: multiply the two signed numbers. The RIGHT part is **negative**.
4. Subtract the absolute value of the right from LEFT $\times 100$.

Examples: - $102 \times 98 \rightarrow 100 \mid -4 \rightarrow 10000 - 4 = \mathbf{9996}$ - $104 \times 96 \rightarrow 100 \mid -16 \rightarrow 10000 - 16 = \mathbf{9984}$ - $108 \times 95 \rightarrow 103 \mid -40 \rightarrow 10300 - 40 = \mathbf{10260}$

Tip: for $(a+x)(a-y)$ with $a = \text{base}$, the left is $a + x - y$ and the right is $-xy$.
Whenever excess equals deficit, the left is exactly the base and the answer is $\text{base}^2 - xy$.

Homework

- 10 mixed problems (mix of both-below, both-above, and mixed-sign). Decide the case first.
- Find ONE example in your textbook where mixed-sign Nikhilam would be faster than long multiplication. Bring it.

Differentiation

- **Tier up:** Try 121×79 (further from 100). Excess $+21$, deficit -21 . Cross: $121 - 21 = 100$. Right: $+21 \times -21 = -441$. Answer: $10000 - 441 = \mathbf{9559}$. (Equivalently: $(100+21)$

$(100-21) = 100^2 - 21^2 = 10000 - 441$.) Notice you’ve just used the *difference of squares* identity. Connect to algebra.

- **Tier down:** Stay with 102×99 , 103×98 (small differences). Build up the sign care before tackling larger gaps.

Teacher notes

- The mixed case is where most students give up and revert to long multiplication. **Don’t let them.** The sign care is genuinely tricky for one day; by Day 6 it’ll be a non-issue.
- The “tier up” connection to difference-of-squares (a^2-b^2 identity) is a real bridge to algebra. If a student spots it, celebrate.
- Some students will resist the negative right part. Show them the “borrow from left” form (above) as an alternative — both forms give the same answer.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, mixed-sign examples.

Day 6 — Near 1000: Padding the Right Part to 3 Digits

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve 3-digit $\times$ 3-digit multiplications where both factors are within ~ 15 of 1000, correctly padding the right part to 3 digits.

Materials

- Whiteboard.
- Drill sheet (8 problems near 1000).
- Reference card: “deficits from 1000” (a few worked: $998 \rightarrow 2$, $991 \rightarrow 9$, $988 \rightarrow 12$, $1003 \rightarrow +3$, etc.).

Lesson flow

Warm-up (5 min)

- Mixed sign-and-magnitude round: teacher says a number, class says deficit (if below 1000) or excess (if above).
- $1003 \rightarrow +3$. $998 \rightarrow -2$. $991 \rightarrow -9$. $1012 \rightarrow +12$. $988 \rightarrow -12$.

Core (15 min) — Same rule, bigger pad

1. **The rule is unchanged.** Set up the vertical line, find deficits/excesses, cross-subtract or cross-add, multiply the deficits/excesses.
2. **The ONE change: pad the right part to 3 digits.** Base 1000 has 3 zeros. The right part must be exactly 3 digits.

3. **Walk through 998 × 997:**

- Deficits from 1000: 2, 3.
- LEFT: $998 - 3 = 995$. (Or $997 - 2 = 995$. ✓)
- RIGHT: $2 \times 3 = 6$. Pad to 3 digits: 006.
- Answer: **995006**. ✓ (Calculator: $998 \times 997 = 995006$.)

4. **Walk through 1003 × 1004:**

- Excesses: +3, +4.
- LEFT: $1003 + 4 = 1007$.
- RIGHT: $3 \times 4 = 12$. Pad to 3 digits: 012.
- Answer: **1007012**. ✓

5. **The overflow case: 988 × 985:**

- Deficits: 12, 15.
- LEFT: $988 - 15 = 973$. (Or $985 - 12 = 973$. ✓)
- RIGHT: $12 \times 15 = 180$. Pad to 3 digits: 180 (already 3 digits, no overflow).
- Answer: **973180**. ✓

6. **The harder overflow: 985 × 978:**

- Deficits: 15, 22.
- LEFT: $985 - 22 = 963$. (Or $978 - 15 = 963$. ✓)
- RIGHT: $15 \times 22 = 330$. Already 3 digits.
- Answer: **963330**.
- Verify: $985 \times 978 = 963330$. ✓

7. **The hard overflow: 970 × 965:**

- Deficits: 30, 35.
- LEFT: $970 - 35 = 935$.
- RIGHT: $30 \times 35 = 1050$. That's 4 digits — overflow! Carry the leading 1: left = $935 + 1 = 936$; right = 050.
- Answer: **936050**. ✓

8. **Mixed-sign case near 1000: 1002 × 998:**

- Excess +2, deficit -2.
- LEFT: 1000 (either cross).
- RIGHT: $+2 \times -2 = -4$.
- Answer: $1000 \times 1000 - 4 = 999996$. ✓

Activity (20 min) — Drill

Drill sheet:

- | | | |
|----------------------|-----------------------|-----------------------|
| 1. 998×997 | 2. 996×995 | 3. 999×998 |
| 4. 988×985 | 5. 1003×1004 | 6. 1005×1002 |
| 7. 1002×998 | 8. 970×965 | |

- 10 min individual silent.
- 10 min paired check + redo errors.

Answer key (teacher): | Q | Sign | Deficits/excesses | Left | Right (padded) | Answer | |—|—|—|—|
 —|—|—|—| 1 | below | 2, 3 | 995 | 006 | 995006 | | 2 | below | 4, 5 | 991 | 020 | 991020 | | 3 | below |
 1, 2 | 997 | 002 | 997002 | | 4 | below | 12, 15 | 973 | 180 | 973180 | | 5 | above | 3, 4 | 1007 | 012 |
 1007012 | | 6 | above | 5, 2 | 1007 | 010 | 1007010 | | 7 | mixed | +2, -2 | 1000 | -4 | 999996 | | 8 |
 below | 30, 35 | 935+1=936 | 050 | 936050 |

Wrap-up (5 min)

- **Formative quiz Part C (3 min)** — see quizzes/formative.md.
- Preview Day 7: “*Tomorrow we prove why this works. One line of algebra.*”

Student-facing content

Near 1000 — 3-digit pad

The rule is unchanged. The **only** change: pad the right part to **3 digits** (because base 1000 has 3 zeros).

Examples: - $998 \times 997 \rightarrow 995 \mid \mathbf{006} \rightarrow \mathbf{995006}$ - $1003 \times 1004 \rightarrow 1007 \mid \mathbf{012} \rightarrow \mathbf{1007012}$ - $970 \times 965 \rightarrow 935+1=936 \mid \mathbf{050} \rightarrow \mathbf{936050}$ (right part overflowed) - $1002 \times 998 \rightarrow 1000 \mid -4 \rightarrow \mathbf{999996}$ (mixed)

General rule: the right-part pad-width equals the number of zeros in the base.

Homework

- 8 problems on the back of today’s worksheet, mixed near-1000.
- Compute one of them with long multiplication. How much longer did long take?

Differentiation

- **Tier up:** Try 9998×9997 (base 10000). Right part pad = 4 digits. Confirm answer: 99950006.
- **Tier down:** Stay with the easy near-1000 problems (998×997 , 999×998). Build up.

Teacher notes

- The padding rule is the only new thing today. If students have Day 3 fluent, this lesson is mostly drill.
- Forgetting to pad is the most common error. A student who writes $998 \times 997 = 9956$ instead of 995006 has forgotten to pad to 3 digits. The diagnostic: their answer has the wrong number of digits.
- The “general rule” (pad-width = number of zeros in base) is a real generalisation worth boxing on the board. It sets up Day 7’s algebra.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sannamish WA) — Nikhilam multiplication, base-1000 examples.

Day 7 — Why It Works: The Algebra

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will state the algebraic identity behind the Nikhilam method $(B - a)(B - b) = B(B - a - b) + ab$, derive it on paper, and explain in their own words why this matches the procedure.

Materials

- Whiteboard.
- The 16-sūtra list (Microsoft Word - sutras1.pdf) — printed and visible.
- Worksheet: “explain to a friend” prompt + space.

Lesson flow

Warm-up (5 min)

- Quick Nikhilam round on 97×96 (chorus answer: 9312) and 103×104 (10712).
- “You’ve been doing this for a week. Today we prove why it works.”

Core (20 min) — The algebraic proof

1. **Set the variables.** Call the base B ($= 100$, say). Call the two factors $B - a$ and $B - b$, where a and b are the deficits.

For 97×96 : $B = 100$, $a = 3$, $b = 4$. Then $97 = B - a$ and $96 = B - b$.

2. **Multiply directly:**

$$\begin{aligned}(B - a)(B - b) &= B^2 - Ba - Bb + ab \\ &= B^2 - B(a + b) + ab \\ &= B(B - a - b) + ab \\ &= B \cdot \text{LEFT} + \text{RIGHT}\end{aligned}$$

where $\text{LEFT} = B - a - b$ and $\text{RIGHT} = ab$.

3. **Match this to the Nikhilam procedure.**

Algebra	Nikhilam step
$B - a - b$	Cross-subtract: take one number minus the <i>other’s</i> deficit
ab	Multiply the deficits for the RIGHT part

Algebra

Nikhilam step

 $B \cdot \text{LEFT} + \text{RIGHT}$

Concatenate (with the right pad)

4. **Check: cross-subtract gives $B - a - b$.**

- $(B - a) - b = B - a - b$. ✓ ($97 - 4 = 93$.)
- $(B - b) - a = B - a - b$. ✓ (Either cross gives the same.)

5. **Why concatenation works.**

- The “LEFT” digit pattern, placed in the next-higher slot, means we’re multiplying it by B . (E.g. $93|12 = 93 \times 100 + 12 = 9300 + 12 = 9312$.)
- So $93|12$ is exactly $B \cdot \text{LEFT} + \text{RIGHT} = 100 \cdot 93 + 12 = 9312$.
- That’s exactly what the formula said.

6. **The “above the base” case.** Use $(B + x)(B + y)$:

$$\begin{aligned}(B + x)(B + y) &= B^2 + Bx + By + xy \\ &= B(B + x + y) + xy \\ &= B \cdot \text{LEFT} + \text{RIGHT}\end{aligned}$$

Now $\text{LEFT} = B + x + y$ — cross-**add**. $\text{RIGHT} = xy$. Sign flips, procedure mirrors.

7. **The mixed case.** Use $(B + x)(B - y)$:

$$\begin{aligned}(B + x)(B - y) &= B^2 - By + Bx - xy \\ &= B(B + x - y) + (-xy) \\ &= B \cdot \text{LEFT} + \text{RIGHT}\end{aligned}$$

$\text{LEFT} = B + x - y$ (signed cross). $\text{RIGHT} = -xy$ (negative). The negative right part is the sign issue from Day 5.

8. **Why the pad-width = zeros in base.** Because $B \cdot \text{LEFT} + \text{RIGHT}$ requires RIGHT to occupy the slots from 0 up to $B - 1$. If $B = 100$, RIGHT is at most a 2-digit number; if it overflows to 100 or more, the overflow rolls into LEFT . That’s the “carry overflow” rule from Day 3.

Activity (15 min) — Explain to a friend

- Worksheet prompt: > Imagine your Grade 4 cousin asks: “*Why does that Nikhilam trick work? Is it magic?*” In 5–7 sentences, explain why it’s NOT magic. Use the words **base**, **deficit**, and **algebra**.
- 10 min individual writing. 5 min pair-sharing: read your paragraph to your partner; they tell you one part that’s clear and one part that’s unclear.

Wrap-up (5 min)

- Read aloud one strong paragraph (with permission).
- The sūtra connection: write on the board:

“*Nikhilam Navataścaramaṁ Daśataḥ*” — “all from 9 and the last from 10” — is the rule for computing the **deficit** quickly (which we drilled on Day 2). The *multiplication* technique uses that deficit via the algebra above.

- Preview Day 8: “Tomorrow — when is this method actually faster? And when should you use long multiplication?”

Student-facing content

The algebra behind Nikhilam

Let B be the base. Let a and b be the deficits.

$$(B - a)(B - b) = B(B - a - b) + ab$$

- $B - a - b$ is the LEFT part (cross-subtract).
- ab is the RIGHT part (product of deficits).
- The concatenation LEFT | RIGHT equals $B \times \text{LEFT} + \text{RIGHT}$ — which is exactly what the algebra produces.

For excess (both above base): $(B + x)(B + y) = B(B + x + y) + xy$. Cross-ADD; right part is positive.

For mixed: $(B + x)(B - y) = B(B + x - y) + (-xy)$. Negative right part.

The sūtra *Nikhilam Navataścaramaṁ Daśataḥ* names the technique for computing the deficit fast. The multiplication shortcut is the algebraic application.

Homework

- Write a 1-paragraph explanation suitable for a Grade 7 student (your own grade level). Include one worked example with the algebra shown.
- Prove $(B + x)(B - x) = B^2 - x^2$ using the above identity. Find an example where this is faster than long multiplication. (Hint: 101×99 .)

Differentiation

- **Tier up:** Generalise. For three factors $(B - a)(B - b)(B - c)$, expand. Does the Nikhilam pattern extend? (Yes — for three factors near a base, see *Vedic Mathematics Batch1.pdf* extension section.)
- **Tier down:** Skip the formal algebra. Verify the identity by plugging in numbers ($B=100$, $a=3$, $b=4 \rightarrow$ both sides give 9312).

Teacher notes

- The honest framing in one sentence: “*The Sanskrit sūtra names the deficit-computation rule. The algebraic identity is what makes the multiplication trick work, and that identity is just standard FOIL / distributivity.*”
- A few students will object: “*If the algebra is just algebra, why call it Vedic?*” Honest answer: “*Good catch. The procedure has a Sanskrit name and was published in 1965 by Tirthaji. The math itself is older than any name — it’s the distributive property. The Vedic-mathematics framing is contested among historians; we’ll engage with that more at Senior band.*”

- Reinforce: this is *not* a debunking. It's an honesty about what's old (algebra), what's modern (the packaging as 16 sūtras), and what's contested (the Vedic provenance).

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, “why it works” section.
- *Microsoft Word - sutras1.pdf* — list of 16 Tirthaji sūtras.
- Optional teacher reference: *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*.

Day 8 — When To Use It: Nikhilam vs. Long Multiplication

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will articulate the regime where Nikhilam is faster than long multiplication, AND will recognise problems where long multiplication (or *Ūrdhva-tiryagbhyām*, the general “vertically and crosswise” method) is the better choice.

Materials

- Whiteboard, two columns labelled “Use Nikhilam” and “Use long / Ūrdhva.”
- “Decide-or-don’t” card deck — 20 multiplication problems on cards. (Teacher prepares ahead.)
- Reflection slip.

Lesson flow

Warm-up (5 min)

- Quick Nikhilam: 97×96 (9312), 104×103 (10712), 102×98 (9996).
- “Now — would Nikhilam work for 47×53 ? Let’s see.”

Core (15 min) — When it works, when it doesn’t

1. **The honest claim.** Nikhilam multiplication is fast *only* when both factors are close to the same power of 10.
2. **Try a “bad” example: 47×53 :**
 - Base = ? Neither factor is near 10 (too big), near 100 (too far below), or near 50 (a *sub-base*; advanced).
 - If you force base 50: deficits $+(-3)$, $+3$. Left = $47 + 3 = 50$. Right = $(-3)(+3) = -9$. Answer = $50 \times 50 - 9 = 2491$. ✓ Slow because you have to compute $50 \times 50 = 2500 =$ a *multiplication*.

- **Sub-bases work** (Day 8 tier-up), but they're not faster than long multiplication for casual users.
- 3. **A genuinely bad case: 47×89 :**
 - Not near any common base.
 - Use long multiplication. **It's faster.**
- 4. **The decision rule (write on board, boxed):**
Use Nikhilam when:
 - Both factors are within ~ 5 of a power of 10 (10, 100, 1000): trivially fast.
 - Both within ~ 10 – 15 of a power of 10: still faster than long multiplication for most students.**Use long multiplication (or *Ūrdhva*) when:**
 - One factor is “random” (not near a base).
 - The factors are near different bases.
 - Speed doesn't matter (writing out long multiplication is more error-resistant).
- 5. **The contrast with Module 7 ($\times 11$) and Module 9 (subtraction).**
 - Module 7's $\times 11$ trick works *always* when one factor is 11. It's a different niche.
 - Module 9's subtraction-by-complement (Nikhilam again) works when subtracting from a power of 10.
 - Nikhilam multiplication is just one shortcut in a toolkit. Long multiplication remains the workhorse.
- 6. **The other Tirthaji sūtra mentioned today: *Ūrdhva-tiryagbhyāṁ*** (“vertically and crosswise”) is Tirthaji's sūtra #3 — the **general** multiplication procedure. It works for any pair of numbers. We won't drill it in this module; it's a topic of its own. Just notice: *Ūrdhva* covers what Nikhilam can't.

Activity (20 min) — “Decide-or-don't” game

Rules: - Pairs receive a deck of 20 cards. Each card has one multiplication problem. - For each card, the pair must **first decide** in 5 seconds: Nikhilam or long? - Then they solve it. - If they used Nikhilam and the method actually applies: full marks. - If they used long multiplication and Nikhilam would have been visibly faster: half marks. - If they used Nikhilam where it wasn't suitable and got stuck: zero marks (but learning is the lesson).

Sample cards:

97×96	→ Nikhilam (both near 100)
47×53	→ Long (no common base)
998×997	→ Nikhilam (near 1000)
73×84	→ Long
103×104	→ Nikhilam
89×91	→ Long-ish (near 90, but no nice base)
102×99	→ Nikhilam (mixed)
347×28	→ Long
988×994	→ Nikhilam (near 1000, larger deficits but still works)

56×89 → Long
 99×99 → Nikhilam (101×99 would also be nice)
 1003×1004 → Nikhilam
 65×75 → Long
 107×108 → Nikhilam
 1234×9876 → Long (too far apart, too random)
 9999×9998 → Nikhilam (near 10000)
 112×113 → Nikhilam (with carry)
 156×178 → Long
 49×51 → Nikhilam (both near 50 if comfortable; else long)
 997×1003 → Nikhilam (mixed, near 1000)

- 20 cards in 20 minutes. Pairs that finish early swap decks with another pair.

Wrap-up (5 min)

- Class discussion: which cards did pairs most often mis-categorise?
- Insight to leave with: **the method is a tool, not a rule.** Long multiplication is *not* “the wrong way.” Each tool fits a certain shape of problem.
- Preview Day 9: speed sprints + project start.

Student-facing content

When to use Nikhilam — and when not to

Use Nikhilam	Use long multiplication
Both factors within ~10 of 10, 100, 1000, 10000	One factor “random” (not near a base)
Mental math, no pencil	Pencil work, no time pressure
Mixed below/above same base	Factors near <i>different</i> bases

The honest truth: Most multiplication problems in life are NOT Nikhilam-shaped. Long multiplication remains your everyday workhorse.

But when both factors *are* near a base — 97×96 , 1003×1004 — Nikhilam is 5–10× faster.

Tirthaji’s other multiplication sūtra, *Ūrdhva-tiryagbhyām* (“vertically and crosswise”), works for any pair of numbers and is taught in its own module. Nikhilam is the *near-base specialist*; *Ūrdhva* is the *generalist*.

Homework

- Find 5 multiplication problems in your school textbook (any subject). For each, decide: Nikhilam, long, or *Ūrdhva*. Write 1 sentence justifying your choice. Bring tomorrow.

Differentiation

- **Tier up:** Try a sub-base (base 50). Compute 47×48 via base 50: deficits 3, 2. Cross: $47 - 2 = 45$. Multiply by 50: 2250. Add deficit product 6 → 2256. (Real: $47 \times 48 = 2256$. ✓) Compare with long multiplication. Which felt faster?

- **Tier down:** Stick to clear-cut cases. Distinguish: “is this near 100?” If yes, Nikhilam. If no, long.

Teacher notes

- This is the honesty day. Some students will be disappointed that Nikhilam isn’t universally faster. That’s the point: claims about Vedic mathematics often overstate the universal advantage. The honest framing protects against later disillusionment.
- The “decide” step before solving is the habit-of-mind worth building. Without it, students apply Nikhilam to inappropriate problems and get slower than the standard algorithm.
- The mention of *Ūrdhva-tiryagbhyām* sets up future modules. Don’t elaborate on the procedure today — it’s a 2-week topic of its own.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam scope and limits.
- *Microsoft Word - sutras1.pdf* — sūtras #2 (Nikhilam) and #3 (*Ūrdhva-tiryagbhyām*).

Day 9 — Speed Sprints + Project Start

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate sub-10-second mental Nikhilam multiplication on Day 1’s baseline problem (and similar) AND will have a project plan/draft ready for Day 10’s showcase.

Materials

- Day 1 baseline sheet (per student) — they pull it from their workbook.
- Sprint sheet: 10 Nikhilam-suitable problems, timer-friendly.
- Project planning slip.
- `activities/activity-02-speed-sprint.md` worksheet.

Lesson flow

Warm-up (5 min)

- Complement-to-100 + excess-from-100 mixed round.
- “*Today is your retest day. Get sharp.*”

Speed sprint (15 min) — `activities/activity-02-speed-sprint.md`

Brief: - 10 Nikhilam problems on a sheet, mixed (both-below, both-above, mixed-sign, near-1000). - 3 minutes individual. Solve as many as you can. - 1 minute self-correction. - Repeat (round 2): swap problems on a fresh sheet, 3 minutes again. - 1 minute self-correction. - 3-minute paired chorus round: partner reads, the other answers in chorus.

Day 1 retest (5 min)

- Students pull their Day 1 baseline sheet ($97 \times 96 = 9312$).
- On a fresh strip, redo the problem using Nikhilam.
- Record new time.
- Compare: **Day 1 (long) vs. Day 9 (Nikhilam)**.

Most students will go from 30–60 sec → under 10 sec. Some will hit 3 sec.

Project planning (15 min)

- Distribute project planning slip (see `assessments/project-brief.md`).
- Each pair fills out:
 - Format chosen (video / workbook / data project).
 - One worked example they'll feature.
 - Materials needed.
 - Who does what.
- Teacher circulates, gives quick approval on viability.

Wrap-up (5 min)

- Brief share-out: 3 pairs announce their plan in one sentence.
- Reminder: project must demonstrate the *Nikhilam* method correctly; must cite at least one source PDF.

Student-facing content

Day 1 retest:

My time on 97×96 (Day 1, long multiplication): ____ seconds My time on 97×96 (Day 9, Nikhilam): ____ seconds **My speedup:** ____×

Sprint scoring: 10 problems in 3 minutes. Each correct = 1 point. Each wrong with right method = 0.5 (you knew the method). Each wrong with wrong method choice = 0.

Project plan (fill in): - Format: _____ - Featured example: _____
- Materials: _____ - Partner / role split: _____

Homework

- Finish project plan at home.
- Practise your featured example until you can present it without notes.
- 5 mental Nikhilam problems a day, every day, even after this module ends.

Differentiation

- **Tier up:** Sprint round of 20 problems in 3 minutes (full speed).
- **Tier down:** Sprint sheet with 5 problems in 3 minutes (slower pace, still timed).

Teacher notes

- The retest moment is the most powerful one of the module. Make space for it — even quiet students will smile when they see their improvement.
- Some pairs won't have a clear project plan by end of class. Pull them aside for 5 min after — better to spend the time now than have a shaky project on Day 10.
- Confirm the data-project pairs have at least 5 willing test subjects lined up.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* — sprint problem source.
- See `assessments/project-brief.md` for the full project specification.

Day 10 — Final Assessment + Project Showcase

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate mastery of the Nikhilam multiplication method via a summative quiz and a 3-minute project presentation.

Materials

- Summative quiz (`quizzes/summative.md`).
- Stopwatch.
- Rubric per pair (`assessments/rubric.md`).
- Reflection slip per student.

Lesson flow

Warm-up (2 min)

- Brief: “Quiz first (15 min), presentations after.”

Part 1: Summative quiz (15 min)

- Distribute `quizzes/summative.md`. Closed notebook.
- Quiz covers:
 - Complement / excess fluency (timed).
 - Both-below-100 (2 problems).
 - Both-above-100 (1 problem).
 - Mixed-sign (1 problem).
 - Near-1000 (1 problem).
 - Decide-which-method (3 problems).
 - One short-answer “why does it work” (algebra).
- Collect at 15-min mark.

Part 2: Project presentations (25 min)

- 3 min presentation + 1 min Q&A per pair (4 min total).
- ~6 pairs fit in 25 min. **Reserve next class for remaining pairs** if needed.
- During each:
 - Teacher marks the rubric in real-time.
 - One assigned audience member asks one question.
 - Other audience members fill “I learned / I wonder” slips.

Wrap-up (3 min)

- Closing question (oral): *“What’s one thing you’ll keep using from Nikhilam? And one thing you’ll keep using long multiplication for?”*
- Reflection slip submitted.

Student-facing content

Today: quiz + presentations

15 min for the summative quiz (closed book).

3-minute presentation per pair, then 1-min audience question.

Reflection (due at end of class): 1. What did you set out to make/learn, and what did you actually end up with? 2. What’s the most surprising thing you discovered while making this? 3. Where will you use Nikhilam in your life outside this classroom?

Homework

- None. Module is complete.

Differentiation

- **Tier up:** Students finishing the quiz early can write a 1-paragraph “what would Module 11 look like for Grade 4?” — input for Primary band.
- **Tier down:** Open-notebook quiz for students with documented learning needs.

Teacher notes

- Mark quiz immediately after class while presentations are fresh.
- If presentations spill into Day 11, protect that time.
- The “reflection on Nikhilam in life” question (oral) — listen for genuine answers (“I’ll use it on price tags in the shop”; “I’ll use it for mental estimation”). The honest answer “I won’t use it much” is also valid.

Citation(s) used in this lesson

- All sources verified across days 1–9.

Module 11 (Middle) — Formative Quizzes

Three short parts: Part A on Day 3 (both below), Part B on Day 4 (both above), Part C on Day 6 (near 1000).

Part A — Day 3 (3 min, 5 problems, both below 100)

Name: _____ Class: _____

Solve each using Nikhilam. Show deficits, left, and (padded) right.

1. $97 \times 96 = \underline{\hspace{1cm}}$ 2. $99 \times 98 = \underline{\hspace{1cm}}$ 3. $94 \times 92 = \underline{\hspace{1cm}}$ 4. $88 \times 89 = \underline{\hspace{1cm}}$ (*watch the overflow!*) 5. $95 \times 95 = \underline{\hspace{1cm}}$

Part A answer key

Q	Deficits	Left	Right (padded)	Answer
1	3, 4	93	12	9312
2	1, 2	97	02	9702
3	6, 8	86	48	8648
4	12, 11	78 (77 + carry 1)	32	7832
5	5, 5	90	25	9025

5 marks (1 each). If a student gets the answer but no working, 0.5 marks; if wrong but method shown, 0.5 marks.

Part B — Day 4 (3 min, 5 problems, both above 100)

Name: _____ Class: _____

1. $103 \times 104 = \underline{\hspace{1cm}}$ 2. $102 \times 107 = \underline{\hspace{1cm}}$ 3. $109 \times 108 = \underline{\hspace{1cm}}$ 4. $112 \times 113 = \underline{\hspace{1cm}}$ (*watch the overflow!*) 5. $105 \times 106 = \underline{\hspace{1cm}}$

Part B answer key

Q	Excesses	Left	Right (padded)	Answer
1	3, 4	107	12	10712
2	2, 7	109	14	10914
3	9, 8	117	72	11772
4	12, 13	126 (125 + carry 1)	56	12656

Q	Excesses	Left	Right (padded)	Answer
5	5, 6	111	30	11130

5 marks.

Part C — Day 6 (5 min, 4 problems mixed-bag + 1 short-answer)

Name: _____ **Class:** _____

1. (1 mark) $998 \times 997 = \underline{\hspace{2cm}}$ 2. (1 mark) $1003 \times 1004 = \underline{\hspace{2cm}}$ 3. (1 mark, mixed-sign) $102 \times 98 = \underline{\hspace{2cm}}$ 4. (1 mark, harder) $985 \times 978 = \underline{\hspace{2cm}}$ 5. (2 marks, short answer) Why does the Nikhilam method work? In one sentence, use the word **base**.

Part C answer key

Q	Answer
1	995006
2	1007012
3	9996
4	963330
5	Because $(B - a)(B - b) = B(B - a - b) + ab$, where B is the base — so cross-subtract gives $B - a - b$ (the left), and the product of deficits gives ab (the right). 2 marks for full reference to algebra; 1 mark if it mentions “base” and “deficits” without the full identity.

6 marks total.

What to do with the results

- **Class average below 60%** (Part A or B) → re-teach the carry-overflow case before moving to next phase.
- **Individual below 40%** → 10-min reteaching block on Day 5 or Day 9.
- **Class average above 80% on Part B** → consider compressing Days 5+6 and using the saved time for Day 9 sprint.

Module 11 (Middle) — Summative Quiz

Time: 15 min · **Marks:** 25 · **Closed notebook**

Instructions: Name + class at top. Show deficits and left/right working. No calculators.

Name: _____ Class: _____

Section A — Complement / Excess Fluency ($5 \times 0.5 = 2.5$ marks, 1 min)

Fill in: deficit (if below) or excess (if above):

97 $\rightarrow$ ____ 88 $\rightarrow$ ____ 103 $\rightarrow$ ____ 998 $\rightarrow$ ____ 112 $\rightarrow$ ____

Section B — Both Below 100 ($2 \times 2 = 4$ marks)

Show working.

1. $96 \times 95 =$ _____

2. $89 \times 87 =$ _____ (*watch the overflow*)

Section C — Both Above 100 ($1 \times 2 = 2$ marks)

3. $107 \times 106 =$ _____

Section D — Mixed Sign ($1 \times 2 = 2$ marks)

4. $103 \times 97 =$ _____

Section E — Near 1000 ($1 \times 3 = 3$ marks)

5. $997 \times 996 =$ _____

Section F — Decide and Solve ($3 \times 1.5 = 4.5$ marks)

For each, write “Nikhilam” or “Long” in the box, then solve.

Problem	Method	Answer
97×102		
47×53		
1003×1002		

Section G — Open Response ($1 \times 7 = 7$ marks)

6. Imagine a Grade 5 student says: “*The Nikhilam method must be magic. How does multiplying small numbers (3 and 4) somehow tell you the answer to multiplying big numbers (97 and 96)?*” In 5–7 sentences, explain why it’s NOT magic. Use the words **base**, **deficit**, and **algebra**. Include the algebraic identity.

Answer key

Q	Answer	Marks
Section A	3, 12, +3, 2, +12	2.5 (0.5 each)
1	9120 (deficits 4,5; left 91; right 20)	2
2	7743 (deficits 11,13; right=143; carry 1; left 77 $\rightarrow$ 78; right 43; final 7743). Calculator confirms $89 \times 87 = 7743$	2
3	11342 (excesses 7,6; left 113; right 42)	2
4	9991 (excess +3, deficit -3; left 100; right -9 $\rightarrow$ $10000 - 9 = 9991$)	2
5	993012 (deficits 3,4; left 993; right 012; $\rightarrow$ 993012)	3
6a	Nikhilam ($102 + 97$ are both near 100): mixed sign. Excess +2, deficit -3. Left=99 or $100 - \dots$ carefully: $102 + (-3) = 99$ or $97 + 2 = 99$. Right = $(+2)(-3) = -6$. Answer = $99 \times 100 + (-6) = 9900 - 6 = \mathbf{9894}$	1.5
6b	Long: $47 \times 53 = \mathbf{2491}$. (Nikhilam doesn’t apply cleanly; no nearby power of 10.)	1.5
6c	Nikhilam: excesses +3, +2. Left = 1005. Right = 006. Answer = 1005006	1.5

Q	Answer	Marks
6 (open)	Look for: (1) explicit reference to $(B-a)(B-b) = B(B-a-b) + ab$ or equivalent; (2) “base” used correctly; (3) “deficit” used correctly; (4) coherent explanation. 7 marks: 3 for the identity, 2 for correct vocabulary, 2 for clarity.	7

Total: 25 marks.

Grading bands (project combined → out of 40): - 32–40: Exceeds - 24–31: Meets - 16–23: Approaches - 0–15: Beginning

Homework

Homework — Module 11 Nikhilam Multiplication (Middle)

Day 1 → Day 2

Task: Practice the deficit-from-10 reflex. Write the deficit of each from 10: $7 \rightarrow \underline{\quad}$; $8 \rightarrow \underline{\quad}$; $9 \rightarrow \underline{\quad}$; $6 \rightarrow \underline{\quad}$; $5 \rightarrow \underline{\quad}$; $4 \rightarrow \underline{\quad}$; $3 \rightarrow \underline{\quad}$; $2 \rightarrow \underline{\quad}$; $1 \rightarrow \underline{\quad}$.

Do all 9 in under 10 seconds. Repeat 3 times.

Time: 5 min.

Day 2 → Day 3

Task: Solve 10 single-digit $\times$ single-digit problems (both 6–9) using Nikhilam. Self-check against your times table. Specifically: 7×8 , 8×9 , 9×9 , 6×7 , 6×9 , 7×7 , 8×8 , 6×8 , 9×7 , 9×8 .

Time: 10 min.

Day 3 → Day 4

Task: Solve 10 problems where both factors are between 85 and 99. Write deficits, left, padded right, and answer for each.

Time: 15 min.

Day 4 → Day 5

Task: Mixed sheet — 10 problems, half both-below-100 and half both-above-100. Decide the case before solving.

Time: 15 min.

Day 5 → Day 6

Task: Bring a list of any 5 multiplication problems from your daily life (price tags in a shop, distances, anything). For each, decide: Nikhilaam-suitable or not? Justify in 1 sentence.

Time: 10 min.

Day 6 → Day 7

Task: Submit your project plan paragraph in writing (4–6 sentences). Choose format. Choose featured example.

Time: 15 min.

Day 7 → Day 8

Task: Write a 1-paragraph explanation suitable for a Grade 7 student of the algebraic identity $(B-a)(B-b) = B(B-a-b) + ab$. Include one worked example with the algebra shown.

Time: 15 min.

Day 8 → Day 9

Task: Practise your featured project example until you can present it without notes. Bring ALL project materials tomorrow.

Time: 10 min.

Day 9 → Day 10

Task: Final touches on your project. Make sure you have: the artefact, your script/notes, your 3-min presentation timed in your head, your citation visible on the artefact.

Time: 15 min.

Writing Prompts

Writing Prompts — Module 11 Nikhilaam Multiplication (Middle)

Six prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

1. (Day 1, end of class, 3 min) When you saw the teacher solve 97×96 in three seconds, what did you feel? Curiosity, suspicion, “that can’t be right”? Be honest — describe the feeling.

2. (Day 9, end of class, 5 min) Compare your Day 1 time on 97×96 (long multiplication) with your Day 9 time (Nikhilam). Has your *feeling* about mental math changed in the last two weeks? Be honest — even “no” is a valid answer.

Explanatory prompts

3. (Day 3 or Day 6, homework, ~120 words) Explain the Nikhilam method to a Grade 4 student who has never seen it. No more than 6 sentences. Use one specific example. Don’t say “magic” — say what’s actually happening.

4. (Day 7, homework, ~150 words) Explain the algebraic identity $(B - a)(B - b) = B(B - a - b) + ab$ in your own words. Use one numerical example. Show how each piece of the identity matches a step in the Nikhilam procedure.

Argumentative prompt

5. (Day 8 or homework, 1 paragraph) Some say “Vedic mathematics” should be a mandatory subject in every Indian school. Others say it should be taught alongside the standard algorithm with honest historical framing. Pick a position and defend it in 5–7 sentences. Use what you’ve learned about (a) when Nikhilam is faster, (b) when long multiplication is faster, and (c) the honest history of Tirthaji’s 1965 book.

Creative prompt

6. (Day 5 or Day 8, optional, 1 page) Write a short script (50–150 words) in which a frustrated student and a confident Nikhilam user have a conversation. The frustrated student raises one good objection to the Nikhilam method (“but most multiplication problems aren’t near 100!”). The confident user gives a clear, honest answer. (No “and then they magically agreed” endings — let the disagreement be real.)

Prompt-to-band notes

- For **Primary** band, prompt 1 works as a drawing or oral response.
- For **Senior** band, prompt 5 deserves a full essay treatment with citations from `sources.md`.

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the rubric — does the writing acknowledge the other side, give a specific example, reach a clear position?
- Prompt 6: no grading. Read the best ones aloud (with permission).

Activity 01 — Mixed-Sign Drill

Module: Nikhilam Multiplication · **Band:** Middle · **Day:** 5 · **Time:** 20 min

Type

Paired practice with self-correction.

Goal

Each student solves 8 mixed-sign Nikhilam problems (one factor above 100, one below), with the partner checking accuracy.

Materials per pair

- 1 printed problem sheet (8 problems below).
- 1 pen each.
- 1 stopwatch (optional — for self-pacing).

Set-up (teacher, before class)

- Print the problem sheet.
- Have the answer key on a separate slip for the pair to self-check after they finish.

Instructions (student-facing handout)

Mixed-Sign Drill — work in pairs.

1. Solo (8 min): each partner solves all 8 problems individually. Don't peek.
2. Pair check (7 min): swap sheets. Partner A reads problem 1 aloud; partner B resolves it from scratch on a fresh strip. Compare answers. If they differ, work out which is right.
3. Last 5 min: each student redoes any 3 problems they got wrong, silently.

Problem sheet

- | | | | |
|----|-----------------|----|-----------------|
| 1. | 102×98 | 5. | 97×104 |
| 2. | 104×97 | 6. | 105×95 |
| 3. | 103×96 | 7. | 101×99 |
| 4. | 106×95 | 8. | 108×92 |

Answer key (teacher copy — DON'T print on student sheet)

Q	Signed parts	Left	Right (signed)	Answer
1	+2, -2	100	-4	9996
2	+4, -3	101	-12	10088
3	+3, -4	99	-12	9888
4	+6, -5	101	-30	10070
5	-3, +4	101	-12	10088

Q	Signed parts	Left	Right (signed)	Answer
6	+5, -5	100	-25	9975
7	+1, -1	100	-1	9999
8	+8, -8	100	-64	9936

Discussion prompts (last 2 min)

- Which problem took the longest? Why?
- Did you notice that Q7 (101×99) is exactly $(100+1)(100-1) = 100^2 - 1^2 = 9999$? That's the difference-of-squares identity. Nikhilam IS that identity for the special case.
- When excess equals deficit (Q1, Q6, Q7, Q8), the left is always exactly the base. Did you spot this shortcut?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Partners writing deficits clearly, debating sign carefully	"Wait, $+4 \times -3$ is -12 , not $+12$."
Needs intervention	One partner just copies the other; or both ignore signs	"I got 10100, you got 10088 — which is it?" → great teachable moment

Differentiation

- **Tier up:** Try 121×79 (large mixed gap; difference of squares — answer 9559). Try 1003×997 (mixed, near 1000 — answer 999991).
- **Tier down:** Stay with 102×99 , 103×98 , 101×99 (small differences) until the sign care is automatic.

Safety / sensitivity

- No timed competition between pairs publicly. Pairs are self-pacing.

Clean-up

- Worksheets collected (for the teacher to spot-check before Day 6).

Activity 1 — Method Speed Race

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Middle · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Multiplication Near a Power of 10 — Nikhilam through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

20 problem cards, stopwatches, two columns on whiteboard

What students do

Split class in two: one team solves with traditional long multiplication, the other with Nikhilam. Race on identical problems. Discuss which problems favour which method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Speed Sprint

Module: Nikhilam Multiplication · **Band:** Middle · **Day:** 9 · **Time:** 18 min within Day 9

Type

Timed individual drill + paired chorus round.

Goal

Each student solves 10 mixed-case Nikhilam problems mentally in 3 minutes; then pairs do a chorus-round of 10 more.

Materials per pair

- 1 sprint sheet (10 problems).
- 1 pen each.

- 1 stopwatch (or use the wall clock).

The sprint problem sheet (round 1)

- | | |
|---------------------|------------------------|
| 1. 97×98 | 6. 88×89 |
| 2. 103×104 | 7. 102×99 |
| 3. 104×105 | 8. 99×99 |
| 4. 105×95 | 9. 998×997 |
| 5. 96×96 | 10. 1004×1003 |

The sprint problem sheet (round 2, alternate)

- | | |
|---------------------|------------------------|
| 1. 95×94 | 6. 89×92 |
| 2. 107×106 | 7. 103×96 |
| 3. 108×104 | 8. 101×101 |
| 4. 106×94 | 9. 997×998 |
| 5. 93×97 | 10. 1005×1004 |

Instructions

Speed Sprint — individual, 3 minutes per round.

1. **Round 1 (3 min):** Solve all 10 problems on sheet 1 as fast as possible. Mental, not paper-and-pencil if you can.
2. **Self-correct (1 min):** Use the answer key.
3. **Round 2 (3 min):** Solve all 10 on sheet 2.
4. **Self-correct (1 min).**
5. **Chorus round (5 min):** Pair up. Partner A reads problem; you both shout the answer. Switch. Cover 10 random problems.

Round 1 answers

Q	Answer
1	9506
2	10712
3	10920
4	9975
5	9216
6	7832
7	10098
8	9801
9	995006
10	1007012

Round 2 answers

Q	Answer
1	8930

Q	Answer
2	11342
3	11232
4	9964
5	9021
6	8188
7	9888
8	10201
9	995006
10	1009020

Scoring

- 10 correct in 3 min = “Exceeds” (sub-20-sec average).
- 7–9 correct = “Meets.”
- 4–6 correct = “Approaches” — practise more before Day 10’s quiz.
- 0–3 correct = “Beginning” — get a teacher 5-min reteach on Day 10.

Going well / Needs intervention

	Looks like	Sounds like
Going well	Quiet sustained focus, fast pencil-strokes, deficit-then-answer	(Nothing — they’re solving)
Stuck	Confused face, restarting same problem 3 times	“Wait, is 88 above or below 100?”

Differentiation

- **Tier up:** Sprint round of 20 problems in 3 minutes; or include 4-digit problems near 10000.
- **Tier down:** 5 problems in 3 minutes (gentler pace); focus on accuracy over speed.

Clean-up

- Sheet collected (for retest record).
- Pair-chorus round is unscored — just for chorus energy.

Project Brief

Project Brief — Module 11 Nikhilam Multiplication (Middle)

Work sessions: Day 9 (planning) + home work · **Presentation:** Day 10 · **Total marks:** 40 (project 30 + presentation 10)

The project

Each pair (or individual, with teacher approval) chooses ONE of three formats:

Format 1: 60-second teaching video

Make a short video that teaches Nikhilam multiplication to a Grade 4 or 5 student. Constraints: - Total length: 60 seconds (± 10 sec). - Must show: **at least one** worked example clearly (deficits, cross-subtract, right-part with pad, final answer). - Bonus: include both a both-below-100 example AND a both-above-100 example. - Cite the source (PDF + sūtra name) in the title slide or closing credit.

Format 2: Printed workbook

Produce a 20-problem workbook for a Grade 4 or 5 self-learner with Nikhilam multiplication. Constraints: - 20 problems organised in 4 difficulty levels (5 problems each). - Level 1: single-digit $\times$ single-digit (base 10). - Level 2: both below 100 (no overflow). - Level 3: both below or both above 100 (mix). - Level 4: mixed-sign + near-1000. - Each level includes ONE fully worked example. - Answer key on the back page. - Title page with cited source (PDF + sūtra name). - Optional: a 1-page “when NOT to use this” note (from Day 8).

Format 3: Comparative speed-test data project

Time 5–10 people (family, schoolmates) on a small set of multiplication problems, comparing long multiplication vs. Nikhilam. Constraints: - 4–6 problems: a mix of Nikhilam-suitable (e.g. 97×96 , 103×104 , 998×997) and Nikhilam-unsuitable (e.g. 47×53). - A table of names (or codes), problem solved, time using standard, time using Nikhilam (if applicable). - A plot or visual comparison. - A 1-paragraph interpretation of the data — including any surprising findings (e.g. “*Nikhilam was only faster when both numbers were near 100; for 47×53 it was slower*”). - Cite the source PDF.

Why this matters

You’re moving from *learner* to *teacher* (or *researcher*). The discipline of explaining or measuring forces you to understand more deeply.

You’ll also learn the meta-lesson: the Nikhilam method is fast in its niche and slow outside it. A good project shows both.

What success looks like

A strong project will: - Demonstrate the Nikhilam method accurately (no arithmetic errors). - Be pitched for the intended audience (a 9-year-old for video and workbook; a peer reader for data). - Show the pair's own thinking. - Be clearly communicated. - Acknowledge the *limit* of the method (when it's faster, when it's not).

Working in pairs

- Pair work strongly encouraged.
- Individual allowed for the workbook or video format.
- Groups of 3 acceptable — each member must have a distinct role.

Citation rules

- At minimum **one** citation must appear in your project — visible.
- Format: <PDF filename> (e.g. *Vedic Mathematics Batch1.pdf*).
- Naming the sūtra (*Nikhilam Navataścaramaṁ Daśataḥ*) is encouraged.
- See `sources.md` for the list of available PDFs.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to learn or make, and what did you actually end up with? 2. What's the most surprising thing you discovered while making this? 3. If you had three more days, what would you do differently?

Rubric

Project Rubric — Module 11 Nikhilam Multiplication (Middle)

Project marks: 30 (out of 40 with presentation 10).

Criteria

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Method demonstrated correctly	Method shown completely (deficits, cross-subtract/add, right-part pad, carry-overflow if applicable); at	Method shown completely; minor errors on 1 example, OR mixed-sign case missing.	Method shown but with a conceptual gap (forgot pad-width, or sign error).	Method incorrect, absent, or applied to a non-Nikhilam problem.	× 2

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	least 3 worked examples spanning both-below, both-above, and (ideally) mixed-sign with no errors.				
Use of sources	Cites at least one PDF section verbatim (e.g. “ <i>Vedic Mathematics Batch1.pdf</i> ”); also names the Sanskrit sūtra <i>Nikhilam Navataścaram aṁ Daśataḥ</i> .	Cites a PDF source.	Mentions a source vaguely.	No source cited.	× 1.5
Audience suitability	Pitched precisely to the target audience (Grade 4 for video; self-study for workbook; readable for data project).	Mostly suitable; minor tonal issues.	Suitable in places, off in others.	Inaccessible for target audience.	× 1.5
Originality	The project shows the pair’s own thinking — a unique example, a custom narrative, a real data set.	Clearly the students’ own work.	Largely repeated from class examples.	Copied verbatim from a source.	× 1
Communication / craft	Visuals, layout, narration are clear, deliberate,	Communicates clearly. 1–2 craft issues.	Message comes through with effort.	Hard to follow.	× 1

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	polished. No arithmetic errors shown.				

Maximum criterion sum (4×7 weights): 28. Project score = (weighted sum) $\times 30/28$.

Presentation (10 marks)

Criterion	4	3	2	1	Weight
Within time (3 min)	2:45–3:00	2:30–3:15	2:15–3:30	Outside	$\times 1$
Both partners contribute	Equal	Both speak unequally	One dominates	One silent	$\times 1$
Clarity	Confident, audible	Audible, mostly clear	Audible only in front	Mostly inaudible	$\times 0.5$

Maximum: 10.

Format-specific notes

- **Video projects:** the 60-second video itself is the artefact; the 3-min presentation explains the design choices (“we used 97×96 because it shows the carry-free case clearly”).
- **Workbook projects:** the workbook is the artefact; the 3-min presentation walks through the cleverest 1–2 problems.
- **Speed-test data projects:** the table/chart is the artefact; the 3-min presentation interprets the data (“our 5 participants showed an average $X\times$ speedup for both-below-100 problems but no speedup for random pairs”).

What “Exceeds” looks like on Method

Look for the *acknowledgement of when not to use Nikhilam*. A project that says “use this for everything!” is missing the Day 8 lesson. A project that says “use this when both are near a power of 10, otherwise use long multiplication” — *that* is Exceeds.